

Contato

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www.linkedin.com/in/tatiane-pereira-dos-santos (LinkedIn)
tatianep.github.io/home (Personal)

Principais competências

Quantum Computing
Quantum Hardware
Device Design

Languages

English (Native or Bilingual)
French (Limited Working)
Spanish (Native or Bilingual)
Portuguese (Native or Bilingual)

Certifications

Introduction to Generative Adversarial Networks (GANs)
Performance Tuning on Apache Spark - Data + AI Summit 2022
What Is Generative AI?
Ethics in the Age of Generative AI
Introduction to Responsible AI

Honors-Awards

Best graduate poster award
1st place awarded with CAPES Exchange Program Scholarship
Awarded with the "FAPERJ Grade 10" Scholarship
"Approved with distinction" in Ph.D. Qualification Exam

Publications

Deep Learning Enabled Strain Mapping of Single-Atom Defects in Two-Dimensional Transition Metal Dichalcogenides with Sub-Picometer Precision
An order N numerical method to efficiently calculate the transport

Tatiane Santos, Ph.D

Scientist | Quantum & AI | Physics PhD
Santa Barbara, Califórnia, Estados Unidos

Resumo

I'm an avid learner, and builder wannabe.

I'm currently working as a Senior Quantum Engineer at Microsoft Station Q, Santa Barbara, CA, applying my unique skill set as a Physicist and AI and Machine Learning Specialist across various industries to help build the first fault-tolerant quantum computer.

Specialties:

- Quantum hardware development
- AI-driven solutions
- Cross-disciplinary collaboration

I thrive at the intersection of science and innovation, and I'm always eager to connect with like-minded professionals exploring the frontiers of technology. Let's build the future together!

Permanent Resident of the U.S.

Experiência

Microsoft
Senior Quantum Engineer
dezembro de 2024 - Present (1 ano 8 meses)
Santa Barbara, California, United States

Working on accelerating the development of quantum hardware with ML, AI and computational physics to optimize qubit performance and automate its development.

ADM
2 anos 2 meses

Data Scientist - Artificial Intelligence
setembro de 2023 - novembro de 2024 (1 ano 3 meses)
Cincinnati Metropolitan Area

properties of large systems: An algorithm optimized for sparse linear solvers

Electron dynamics in extended systems within real-time time-dependent density-functional-theory

The 2021 ultrafast spectroscopic probes of condensed matter roadmap

Distinct Lifetimes for X and Z Loop Measurements in a Majorana Tetron Device

Developed multiple AI chatbots, empowering technical and non technical audience to ask questions to files and datasets, increasing business analysis efficiency.

Built ML solutions for nutrition, supply chain and logistics applications, with impact of up to millions of dollars.

Educated global senior leaders about AI usage and its applications, increasing the impact of AI on productivity throughout the company, from everyday tasks up to custom AI solutions.

Built end-to-end AI and ML solutions using Python, SQL, JavaScript and Azure Machine Learning, leveraging 100% of the tools provided for AI engineering applications within the company.

Data Scientist - Artificial Intelligence - IT Project Manager
outubro de 2022 - setembro de 2023 (1 ano)

São Paulo, Brazil

Worked as the key individual for the collaboration between Brazil and US based teams, unlocking AI initiatives in South America.

Converted web-scraped data, geo-analysis, clustering algorithms, and neural networks models into more efficient marketing strategies and personalized customer experiences.

Wildlife Studios

Associate Data Scientist

dezembro de 2021 - outubro de 2022 (11 meses)

São Paulo, Brazil

Extracted business insights from data models and converted into improved decision-making, with impact in revenue up to 20% at reduced cost of implementation.

Developed advanced business metrics, statistical models, Machine Learning algorithms and Big Data programming strategies, increasing in-app purchases and expanding user base.

Developed several analysis models on a weekly basis, on pricing, engagement, new products performance, anomaly detection, recommendation algorithms, and A/B testing.

University of Illinois at Urbana-Champaign

2 anos 11 meses

Postdoctoral Researcher

janeiro de 2019 - novembro de 2021 (2 anos 11 meses)

Urbana, Illinois, United States

Implemented novel scattering mechanism into C++ suite for improving simulations of NV centers in Diamond, leading to publications.

Performed structure analysis of defects in 2D materials for Machine Learning publication with 100+ citations.

Wrote grants, papers, and scientific reports. Advised Computer Scientist student. Presented in several local and international conferences.

Course Instructor

março de 2020 - junho de 2020 (4 meses)

Urbana, Illinois, United States

Lectured Materials Properties to undergraduate students of UIUC engineering program.

Universidade Federal Fluminense

4 anos 1 mês

Research Assistant

dezembro de 2014 - dezembro de 2018 (4 anos 1 mês)

Niterói, Rio de Janeiro, Brazil

Developed the most precise model of the Quantum Hall Effect in realistic-size samples for the time (100K nanometers squared) using Python and Fortran.

Demonstrated the computational efficiency of linear algebra developed on top of sparse matrix algorithms, leading to publication.

Graduated with the highest GPA of the department, with a 99.9% score.

Graduate Teaching Assistant

julho de 2015 - julho de 2016 (1 ano 1 mês)

Niterói, Rio de Janeiro, Brazil

Lectured Experimental Physics III - Thermodynamics, Fluids, Waves and Optics - to Physics, and Engineering undergraduate programs.

Commissariat a l'Energie Atomique et aux Energies Alternatives Visiting Scientific Researcher

junho de 2017 - dezembro de 2017 (7 meses)

Grenoble, Auvergne-Rhone-Alpes, France

Collaborated with the developers of a quantum dynamics Python package (+100K downloads) on the release of the package's extended version.

Developed high-performance computing Python scripts to simulate particle dynamics in semiconductors and 2D materials, leading to publication.

Universidade Federal do ABC

Research Assistant

setembro de 2012 - setembro de 2014 (2 anos 1 mês)

Santo André, São Paulo, Brazil

Demonstrated chaos theory properties and its implications in classical and quantum dynamics using Fortran simulations.

Universidade Federal do ABC

Teaching Assistant

maio de 2008 - abril de 2012 (4 anos)

Santo André, São Paulo, Brazil

Assisted courses for the Mathematics, Physics, Computer Science and Engineering undergraduate programs and Physics graduate program.

Formação acadêmica

Universidade Federal Fluminense

Ph.D, Physics · (2014 - 2018)

Universidade Federal do ABC

Master's degree, Physics · (2012 - 2014)

Universidade Federal do ABC

Bachelor of Science - BS, Physics · (fevereiro de 2008 - fevereiro de 2012)

Universidade Federal do ABC

Bachelor of Science - BS, Science and Technology · (fevereiro de 2008 - fevereiro de 2012)

Etec Lauro Gomes

Technical degree, Informatics · (2007 - 2008)